Postgraduate Diploma in Data Analytics

Programming for Data Analytics
PDAN8411

Part 1:
Linear regression

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Medical Aid Cost Predictions

Introduction

Organisations utilise predictive analytics to forecast future outcomes using historical data. This study shows how linear regression is used to estimate medical costs based on the user's livelihood, including Age, BMI, region, and smoking habits. The aim is to assist the medical aid company in structuring the cost and improving decision-making. Linear regression was used because of its ability to demonstrate a relationship between a continuous dependent and multiple independent variables.(Kumar $ Singh, 2023). Data analysis process, model development, evaluation, and key findings will be outlined in this report.

1. Data loading and exploratory

Inspection was conducted on the insurance.csv dataset to understand its structure, data type, and completeness, The Dataset contains 1338 records, some variables are categories as "object types" and require conversion. There are no missing values that were found; this indicates the reliability of the data. Descriptive Analytics exposes that medical charges were slightly changed, proving that a small number of customers experienced a very high cost. Analysis indicated that smoking status has a strong impact on charges, with smoking causing higher costs. Age and BMI show a positive correlation with medical cost, while the number of children and region have a weaker influence

1.1. Loading insurance data file and display 5 first rows.

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

Each row represent a client and each column represent a feature used to predict insurance charges.

1.2. Checking the structure of the dataset

Checking the number of rows, columns, column names, datatypes and Non-null values

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1338 entries, 0 to 1337
Data columns (total 7 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   age       1338 non-null   int64
 1   sex       1338 non-null   object
 2   bmi       1338 non-null   float64
 3   children  1338 non-null   int64
 4   smoker    1338 non-null   object
 5   region    1338 non-null   object
 6   charges   1338 non-null   float64
dtypes: float64(2), int64(2), object(3)
memory usage: 73.3+ KB
```

1.3. <u>Checking the missing value</u>

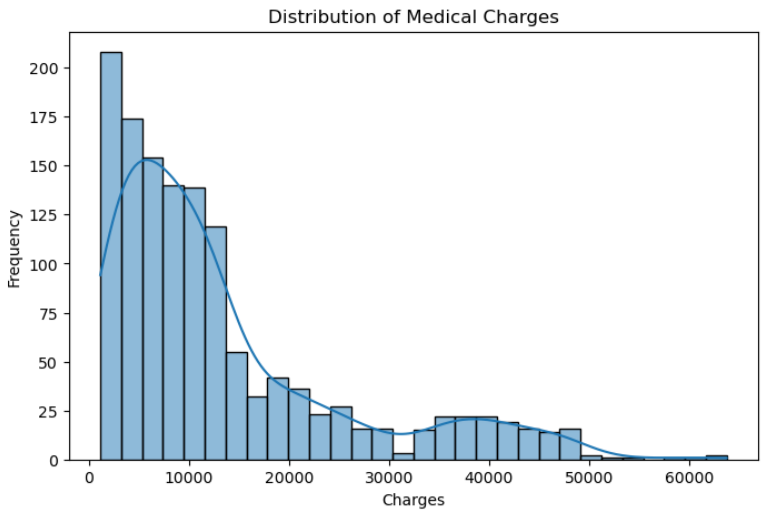
```
[8]:

age         0
sex         0
bmi         0
children    0
smoker      0
region      0
charges     0
dtype: int64
```

.

2. Data Analysis (EDA)

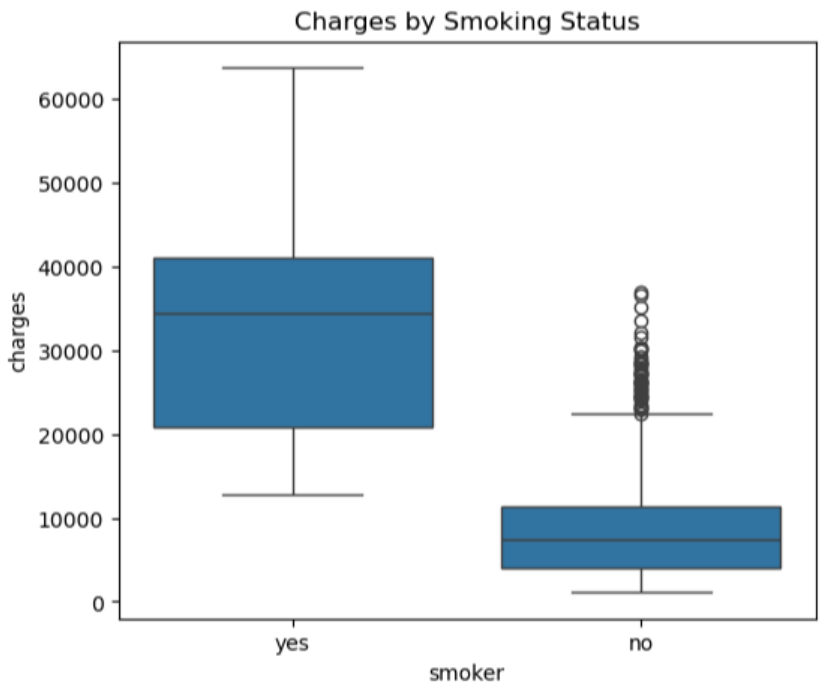
<u>Distribution of Medical Charges</u>

Demonstrates the distribution of medical costs among people. Due to the data's right skew, most people have low expenses, while a small percentage have extremely high ones.



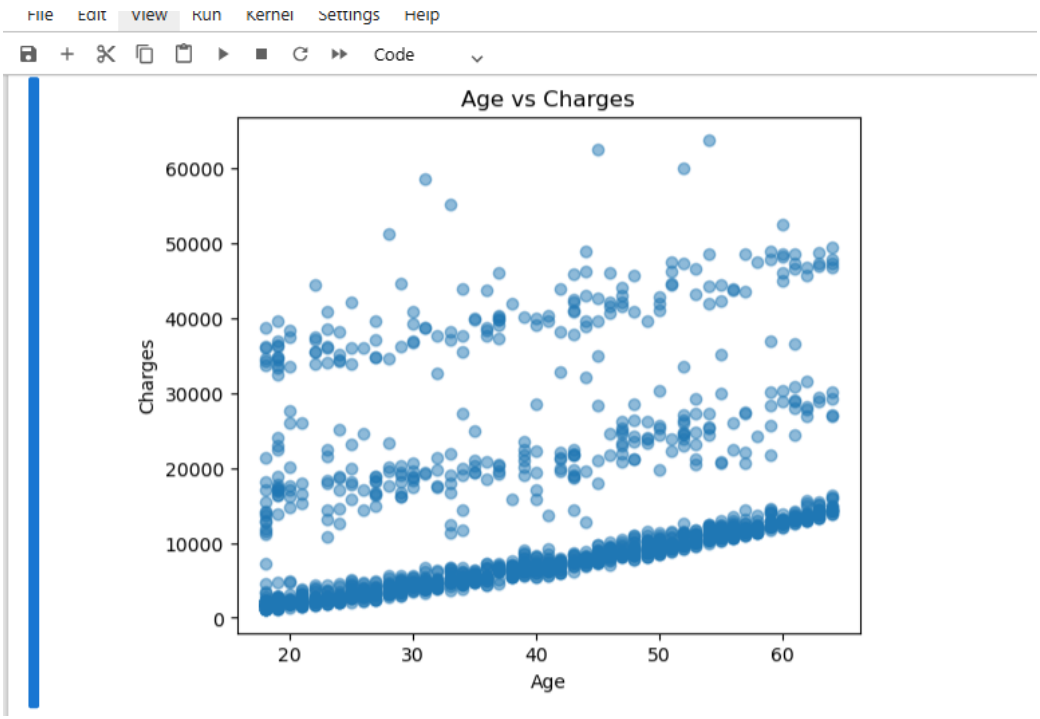
Charges by smoking Status

Compares and contrasts smokers' and non-smokers' medical costs. Smokers' fees are clearly higher and more erratic than those of non-smokers.



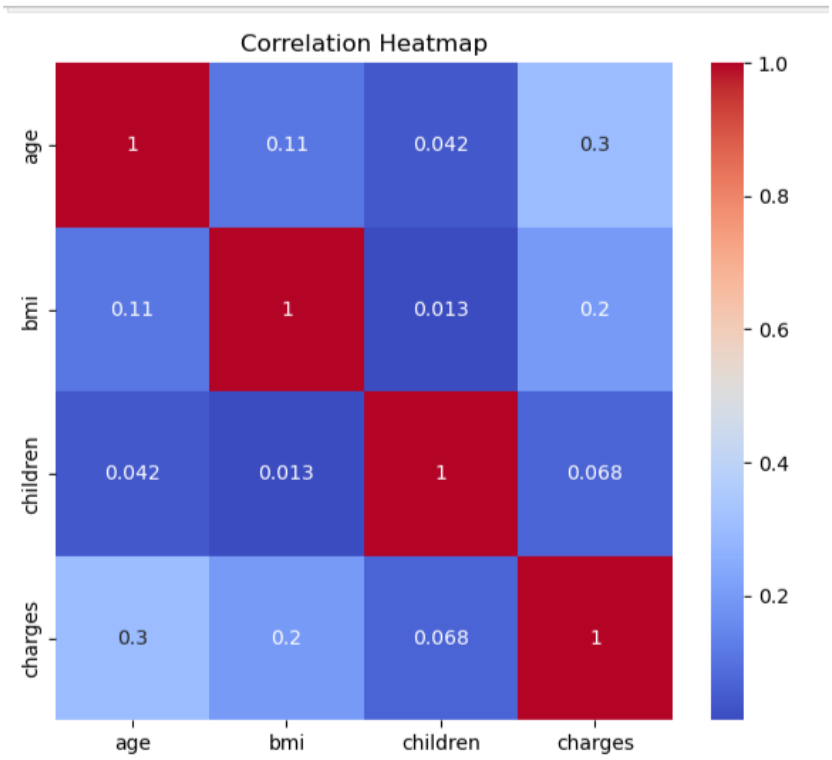
Charges by Age

Demonstrates the connection between medical costs and age. Elderly people typically have higher expenses due to a general upward trend.



Correlational Heatmap

Displays the strength of relationships between variables. Smoking, age, and BMI show the strongest positive correlation with medical charges.



3. Data pre-processing and feature selection

Categorical variables such as sex, smoker, region were converted to a numerical format because a machine learning model requires numerical input. Feature selection was based on correlation analysis and domain relevance; no big features were removed because most variables play a big role in the model. Model accuracy and performance are improved by correct pre-processing (Adivasi & Musa, 2024).

	age	bmi	children	charges	sex_male	smoker_yes	region_northwest	region_southeast	region_southwest
0	19	27.900	0	16884.92400	False	True	False	False	True
1	18	33.770	1	1725.55230	True	False	False	True	False
2	28	33.000	3	4449.46200	True	False	False	True	False
3	33	22.705	0	21984.47061	True	False	True	False	False
4	32	28.880	0	3866.85520	True	False	True	False	False

4. Model training

Model performance was evaluated on unseen data by splitting the dataset into 80% for training and 20% for testing sets. A linear regression model was trained using the training data set. Linear regression works on the assumption that there is a relationship between variables, the errors are independent of each other, and the variance of the errors remains constant (Homoscedasticity).

Train Linear Regression model

```
20]: model = LinearRegression()
     model.fit(X_train, y_train)
```

```
20]: LinearRegression
     ▼ Parameters
```

fit_intercept	True
copy_X	True
tol	1e-06
n_jobs	None
positive	False

5. Model evaluation

5.1. The following metrics were used to evaluate the model:

- **R2 Score:** it was utilised to measure how well the model explains the variance in data.
- **Mean Absolute Error (MAE):** It was used to measure the average prediction error.
- **Root Mean Squared error (RMSE):** It was used to penalize large errors.

5.2. **Results interpretation:**

```
R2 Score:  0.7835929767120724
Mean Absolute Error (MAE):  4181.1944737536505
Root Mean Squared Error (RMSE):  5796.284659276272
```

- The model achieved a relatively high R2 score, showing a good predictive ability.
- MAE and RMSE values suggest moderate prediction error.
- The model performs well overall but struggles with extreme values (High charges)

These results indicate that Linear Regression is effective but may not fully capture complex relationships in data. Model evaluation is important for assessing performance and identifying limitations (Brown & Lee, 2023).

6. **Conclusion**

The analysis proves that lifestyle factors, specifically smoking, influence medical costs. Age and BMI also contribute to increased expenses. While the Linear Regression model provides useful insights. It assumes linear relationships and may not handle non-linear patterns effectively.

Reference

Adeyemi, T., & Musa, A. (2024). *Data preprocessing techniques for machine learning models. Journal of Data Science Applications*, 12(2), 45–60.

Brown, L., & Lee, K. (2023). *Evaluating machine learning models: Metrics and best practices. International Journal of AI Research*, 9(1), 101–115.

Kumar, R., & Singh, P. (2023). *Introduction to predictive analytics using regression models. Data Analytics Review*, 15(3), 67–82.

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